Electric fields

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use field arrows to predict the force on a test charge.

SUCCESS CRITERIA - I can...

1. Draw electric field arrows for isolated positive and negative charges.
2. Use the positive-test-charge convention.
3. Distinguish the field from a particular test charge's force.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use "electric field" and "charge" to explain your model or result. Sentence starters are optional.

electric field: force per positive test charge at a location

charge: a property responsible for electric interactions

vector: a quantity with size and direction

01 Warm-up: bring back an earlier idea

A ray enters a medium with speed 1/3 of its original speed at an angle.
Sketch the bend and state whether frequency changes.

Word help: frequency = number of cycles each second

Sentence starter: The ... in my model represents ... . Its label shows ... .

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: E = F/q; F = qE.
Use consistent signed components. The field points along the force on a positive test charge; negative charge reverses force direction.

Worked reasoning: At a point where E=4 N/C right, a +2 C test charge feels 8 N right. A-2 C charge feels 8 N left. Field direction is defined using positive test charge.

Your turn: Draw the field arrow once and the two alternative force arrows.

Sentence starter: The ... in my model represents ... . Its label shows ... .

Field / first contribution

Force / second contribution

Draw separate arrows; label quantities and units.

Marzano check - circle: ① ② ③

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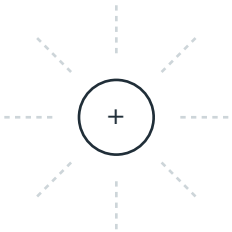
Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Read and interpret the model

Sketch field arrows around a positive source charge. Place a negative test charge to its right and draw that charge's force.

Sentence starter: The ... in my model represents ... . Its label shows ... .



Add arrowheads to show field direction.

The center marks the source; add arrowheads on the surrounding lines to show field direction.

Marzano check - circle: 1 2 3

04 Apply the physics

An electric-field simulation uses a positive 2 C test charge that feels 8 N right.
Find the field magnitude and direction.
Draw separate field and force arrows and label their different units.

Word help: magnitude = size of a quantity without its direction

Sentence starter: Both ... . They differ in ... because ... .

Field / first contribution

Force / second contribution

Draw separate arrows; label quantities and units.

Marzano check - circle: 1 2 3

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Explain your reasoning

If you remove the test charge, does the source charge's field disappear?
Explain the role of a test charge.

Sentence starter: My claim is … . The diagram/data show …, so … .

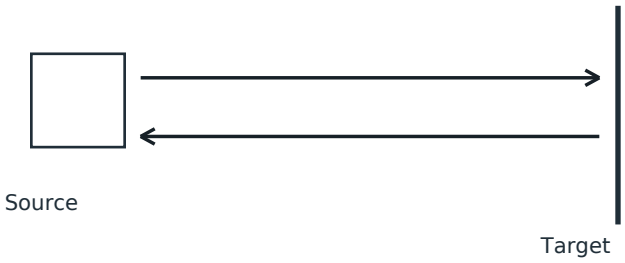
Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

An echo travels 120 m on its full out-and-back path.
Find the wall's distance.
Label the outgoing and returning paths.

Word help: echo = a sound received after reflection

Sentence starter: I found … [unit]. This value means … in this situation.



Marzano check - circle: 1 2 3

07 Exit check

A field points down. Which way is the force on a positive charge? On a negative charge?

Sentence starter: I notice … . This supports … because … .

Field / first contribution

Force / second contribution

Draw separate arrows; label quantities and units.

Marzano check - circle: 1 2 3

08 STRETCH: test the model

Return to the worked example in problem 2.
Choose two different charge values giving forces with equal magnitudes and opposite directions in the same field.

Sentence starter: Both … . They differ in … because … .

Field / first contribution

Force / second contribution

Draw separate arrows; label quantities and units.

Marzano check - circle: 1 2 3